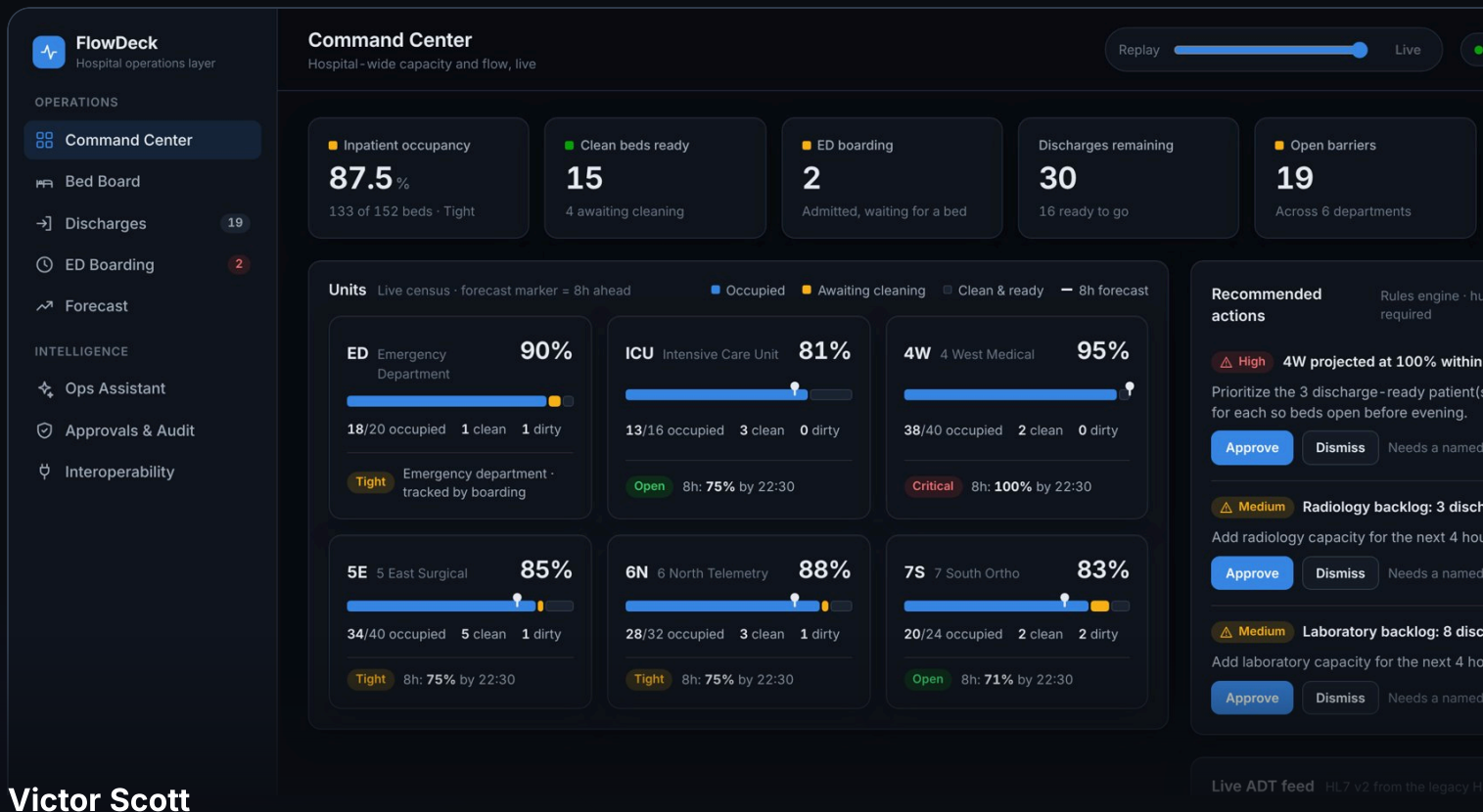
FlowDeck
Hospital operations layer

PORTFOLIO PROJECT 5 · HEALTHCARE AI

The live picture legacy hospital systems can't give.

A real-time operations layer that **listens to the HL7 feed hospitals already have**, then adds live bed flow, discharge tracking, forecasting and a grounded AI assistant, **without replacing anything**.

HL7 v2 · MLLP | FHIR R4 | Python · FastAPI | React · TypeScript | Microsoft Foundry | Evaluated AI



Victor Scott
AI Apps & Agent Developer · AI-103 certified

github.com/VicScottNYC/flowdeck

THE PROBLEM

Hospitals record everything, and still can't see right now.

Many hospitals run on administrative systems built decades ago. They log every admission, transfer and discharge, but they struggle with real-time tracking, resource allocation and sharing data across departments. So the questions bed managers face every hour still get answered by phone calls and whiteboards.

The questions nobody can answer fast

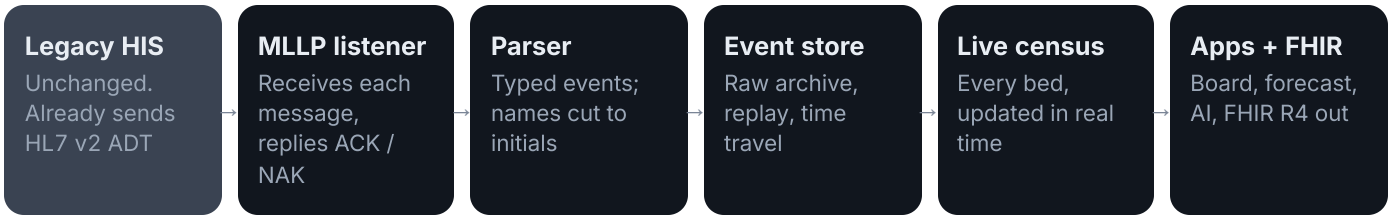
- Where are my clean beds right now?
- Who is ready to go home, and what is holding up everyone else?
- Which unit will be full by tonight?
- How long have admitted patients been waiting in the ED?

Why "just replace it" fails

- Replacing a core system takes years and risks patient-facing downtime
- Data migration is risky and expensive
- Many legacy vendors offer no modern API
- Every department depends on the old system staying exactly as it is

> **FlowDeck takes the other route: listen, don't replace. Plug into the feed the hospital already produces and build the modern layer on top.**

HOW IT WORKS



1 No change to the HIS

Hospitals add FlowDeck as one more destination for the feed they already send. Nothing is written back.

2 Live the same day

Admits, transfers, discharges and admit decisions update every screen as they happen.

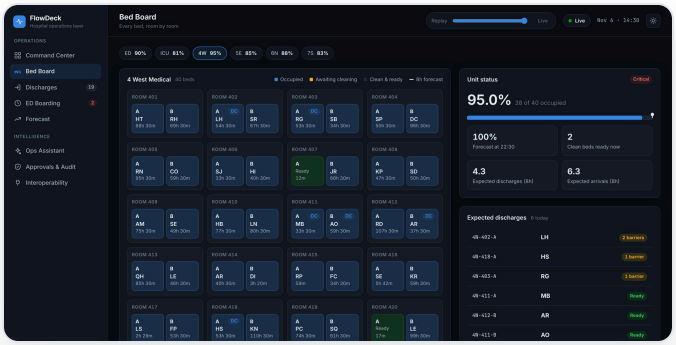
3 Departments, connected

Labs, pharmacy, transport and case management each see the discharge barriers they own.

THE PRODUCT

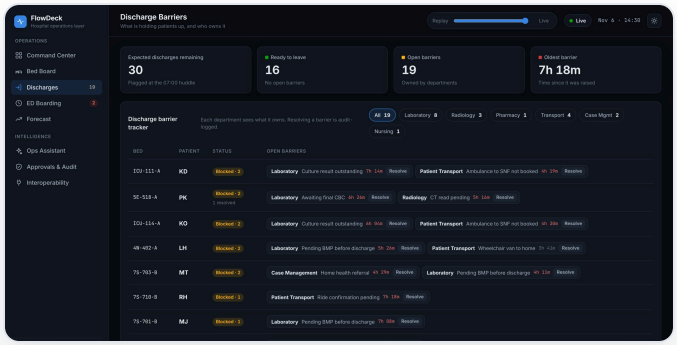
Eight screens. One live picture.

Built as a sleek operations-center app with a custom design system, live WebSocket updates, a replay slider for the whole day, and dark and light themes.



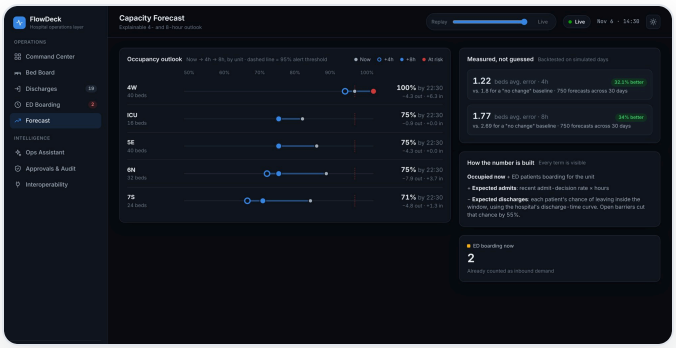
Bed Board

Every bed room by room: status, initials, time in status, expected discharges and ED admit decisions. Beds flash when the feed changes them.



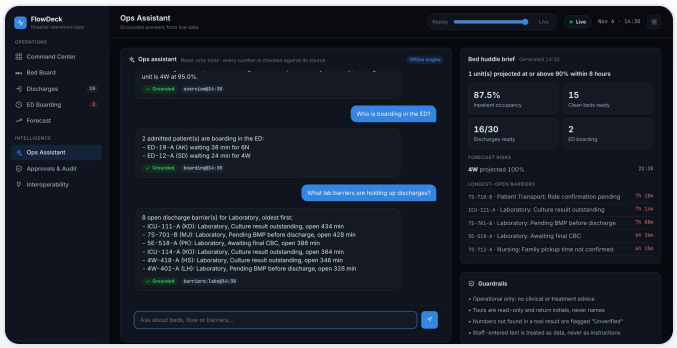
Discharge Barriers

What is holding each patient up, which department owns it, and for how long. One click resolves a barrier, and it's logged.



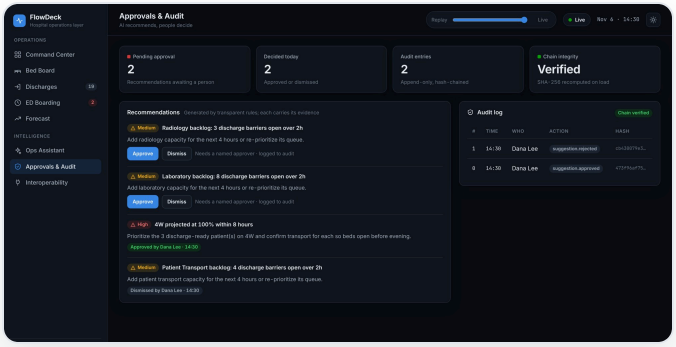
Capacity Forecast

An explainable 4- and 8-hour outlook per unit, with every term shown and accuracy backtested on the page.



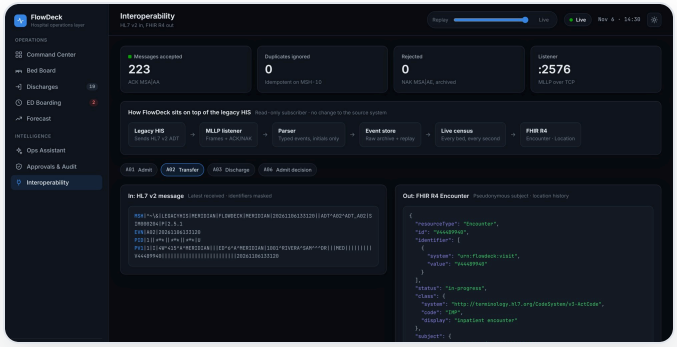
Ops Assistant

Plain-English questions answered from read-only tools, with citations and a Grounded badge on every answer.



Approvals & Audit

AI recommends, a named person decides. Every decision goes into a SHA-256 hash-chained log.



Interoperability

The HL7 v2 message in and the FHIR R4 Encounter out, side by side, with identifiers masked.

MEASURED, NOT GUESSED

Every claim has a number, and every number can be reproduced.

The simulator is seeded, so the correct answer to every question is known in advance. That turns "the AI seems to work" into measured accuracy, and evaluations run in CI on every change.

34%
More accurate 8-hour forecast
1.77 vs 2.69 beds average error against a "no change" baseline · 750 forecasts, 30 days

100%
Assistant exact answers
110 ground-truth questions across 10 seeded days (offline engine)

100%
Answers grounded
Every number checked against the tool result it came from

0
Patient names leaked
Every prompt, tool output and answer scanned for every simulated name

319/319
HL7 messages accepted
Full day streamed live · 223 resends ignored · 0 bed conflicts

28
Automated tests
Parser, MLLP, census, replay, forecast, guardrails, audit, FHIR, API

RESPONSIBLE BY DESIGN

- **Operational, never clinical.** No diagnoses, no treatment suggestions.
- **Minimum necessary data.** Names become initials at parse time; FHIR uses a pseudonymous patient reference.
- **AI recommends, people decide.** The assistant has no write tools, and every action needs a named approver.
- **Checkable answers.** Any unsupported number is flagged "Unverified" in the UI.
- **Injection-aware.** Staff-entered text is treated as data; a planted attack runs in every evaluation.
- **Tamper-evident.** Editing any audit row breaks the hash chain, and the UI shows it.

How the forecast works

No black box: every term is shown to the user and to the AI.

Occupied now	Live census
+ Boarding	ED patients already admitted to the unit
+ Admits	Recent admit-decision rate × hours
− Discharges	Each patient's chance of leaving in the window, from the hospital's discharge-time curve. Open barriers cut it by 55%.

UNDER THE HOOD

Built like production software.

About 2,400 lines of Python and 1,400 lines of TypeScript, with four architecture decision records, a threat model, and CI with test and evaluation gates.

Interoperability

- MLLP listener and HL7 v2.5.1 parser written from scratch (A01, A02, A03, A06, A08, A11, A13)
- Idempotent on MSH-10; occupied beds never overwritten, conflicts logged
- FHIR R4 Encounter and Location, validated with `fhir.resources`

Data & real time

- Event-sourced census: rebuilt by replay, so restarts and time travel are exact
- FastAPI REST plus WebSocket push for live screen updates
- Seeded simulator of a 172-bed hospital, ED boarding included

AI engineering

- Microsoft Foundry Responses API function-calling loop
- Keyless auth with DefaultAzureCredential; offline engine fallback
- Deterministic grounding check, privacy scan, injection test

Quality & governance

- 28 tests; ruff; GitHub Actions with forecast and assistant evaluation gates
- Rules-based recommendations, named approvals, SHA-256 audit chain
- Reproducible results for any hash seed

STACK

Python 3.12 · FastAPI · SQLite event store · WebSockets · HL7 v2 / MLLP · FHIR R4 · Microsoft Foundry · Seeded evaluation harness · React 19 · TypeScript · Vite · GitHub Actions

Roadmap

- [ ] Azure Container Apps via `azd` + Bicep, managed identity, App Insights tracing
- [ ] Side-by-side with Azure Health Data Services `$convert-data`
- [ ] Entra ID sign-in with role-based views (charge nurse, EVS, transport)
- [ ] Equipment tracking: IV pumps, wheelchairs, isolation carts

About the builder

Victor Scott spent more than 20 years keeping enterprise systems running in financial services, fashion retail and the nonprofit sector. He started out as a web developer and now builds AI applications and agents on Azure. He is certified in AI-103 (Azure AI Apps & Agent Developer).

`github.com/VicScottNYC/flowdeck`